A Novel Algorithm for Nighttime Image Dehazing

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Abstract

In recent years, nighttime images collected under haze weather often have problems such as low contrast and serious texture loss. The previous algorithms still have some limitations for nighttime dehazing. In this paper, we proposed a novel nighttime dehazing algorithm and compare our results with existing daytime and nighttime dehazing algorithms' results. The experimental results show that the proposed algorithm has a better dehazing performance and shorter runtime. This algorithm can effectively address image distortion, texture loss and low brightness of the dehazing images.

Author Keywords

nighttime image dehazing; adaptive global brightness compensation; atmospheric scattering model; contrast constraint

1. Introduction

Since aerosol particles can absorb and scatter the light resulting in low-visibility during propagation in haze weather, the obtained images and videos often have blurred visual effects, poor contrast and other problems that bring hidden dangers to the daily management of cities[1, 2]. Along with the rapid development of dehazing technology, research on daytime dehazing algorithms has become increasingly mature. However, there are still great limitations on nighttime dehazing. Nighttime pictures are imaged by point-light sources, which have some problems such as halo interference and over-dark interference. If these degraded images are processed by using daytime dehazing algorithms directly, the results are often below scholars' exception. Therefore, research on nighttime image dehazing has important for image processing and computer vision[3].

In view of the characteristics of degraded nighttime images, Xu et al[4] proposed a nighttime dehazing algorithm based on the full-scale Retinex. They computed the efficient estimate by mapping the transmittance to a surround function, which effectively improves the contrast and the brightness of the degraded image. Nevertheless, further efforts can be made in texture retention. Ding et al[3] demonstrated a nighttime dehazing algorithm based on Retinex and contrast constraint, which achieves dehazing by using constraint function and guided filtering. This approach has an obvious dehazing result and strong edge preservation. However, further improvement in color retention is needed.

To solve the above problems such as image distortion and texture loss, this paper proposed an adaptive global brightness compensation algorithm for nighttime dehazing based on He's dark channel prior algorithm[5, 6]. This approach is physically valid and able to handle complicated illumination conditions at night. It can achieve high-quality dehazing results by using this algorithm.

2. Proposed Method

The atmospheric scattering model has good performance and can be fully used in image dehazing. In this paper, an improved atmospheric scattering model is proposed to reconstruct images without hazy. Firstly, an adaptive global brightness compensation algorithm (AGBC) is used to adjust the illumination of the degraded image $I(x, y)$. Secondly, pseudo-dehazing image $J'(x, y)$ can be obtained by using homomorphic filtering (HF) and contrast limited adaptive histogram equalization (CLAHE). Joint bilateral filtering is mainly used to get the refined transmittance $t''(x, y)$ from the coarse transmittance $t'(x, y)$ which is computed by the atmospheric light value estimation map $A(x, y)$ and the pseudo-dehazing image $J'(x, y)$. Finally, the nighttime dehazing image $J(x, y)$ will be acquired by the proposed model. The overall implementation flow is shown in Figure 1.

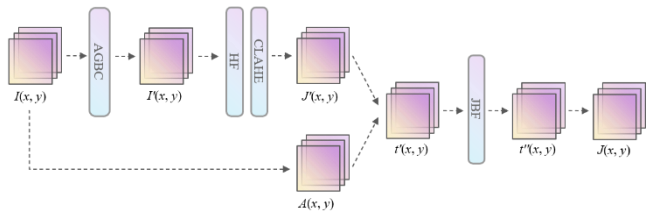

Figure 1. The overall implementation flow

2.1 Adaptive Global Brightness Compensation Algorithm

The proposed algorithm is mainly used to modulate the brightness of nighttime degraded images to obtain a more balanced image with local brightness. Firstly, the nighttime hazy image $I(x, y)$ is converted from the RGB color model to the HSV color model. The brightness image $I_v(x, y)$ takes brightness equalization processing, which is separated from $I(x, y)$. And the process is shown in Figure 2.

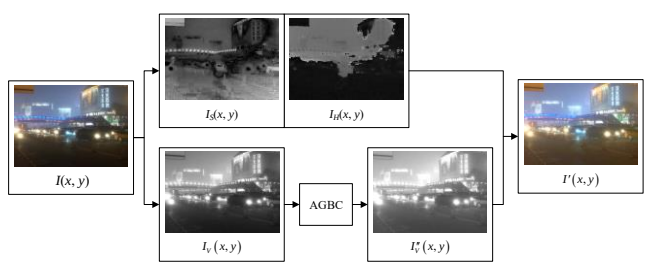

Figure 2. Adaptive Global Brightness Compensation Algorithm

A window of size $n \times n$ is used to compute the average brightness of image $I_v(x, y)$ as Equation (1) shows:

$$\bar{p}_V\left(x_0, y_0\right) = \frac{\sum_{x=x_0, y=y_0}^{x_0+n, y_0+n}\left[p_V\left(x, y\right)\right]}{n^2} \tag{1}$$

where $\bar{p}_V\left(x_0, y_0\right)$ is the average brightness of the image $I_v(x, y)$ and $p_V(x_0, y_0)$ indicates the brightness of the image $I_v(x, y)$ at pixel position (x_0, y_0). Secondly, The brightness image $I_v(x, y)$ is equalized to improve the visibility of the darker regions of the image, and the process is shown in Equation (2):

$$I'_V(x,y) = p_V(x,y)\cdot\left(1+\frac{n^2\cdot\left[p_V(x,y)-\bar{p}_V(x_0,y_0)\right]}{\sqrt{\left[\sum_{i=x,j=y}^{x+n,y+n}\left|p_V(x,y)-\bar{p}_V(x_0,y_0)\right|^2\right]}}\right) \quad (2)$$

where $I'_V(x, y)$ is the brightness image after brightness equalization. In order to make the brightness distribution of image $I'_V(x, y)$ more uniform, adaptive brightness modulation is also required, and the modulation parameter is obtained as Equation (3) shows:

$$\sigma = \frac{n^2\cdot\max_{s\in\Omega_{x,y}(n)}\left[p_V(x,y)\right]}{\left[\sum_{x=x_0,y=y_0}^{x_0+n,y_0+n}\left|p_V(x,x)-\bar{p}_V(x_0,y_0)\right|\right]} \quad (3)$$

where σ is the adaptive brightness modulation parameter and $\Omega_{x,y}(n)$ denotes the maximum filtering window whose size is $n \times n$. According to the parameter σ obtained from Equation (3), this modulation can be accomplished as Equation (4) shows.

$$I''_V(i,j) = (1+\sigma)\cdot p'_V(i,j) \quad (4)$$

where $I''_V(x, y)$ is the final modulation result. Finally, the compensated image $I'(x, y)$ can be obtained while the brightness channel $I''_V(x, y)$, hue channel $I_H(x, y)$ and the saturation channel $I_S(x, y)$ are fused and converted to RGB color model.

2.2 Transmittance Acquisition

The color distortion of the restored image is mostly caused by the inaccurate estimation of transmittance. In order to effectively reduce the color distortion of the restored image, the pseudo-dehazing image is imported as the modulation parameter based on He's transmittance estimation method[5] in this paper.

HF is an image enhancement algorithm, which is easy to implement and low time-consuming; CLAHE can effectively improve the quality of images and reduce the color distortion of images. Therefore, HF and CLAHE are used to process the compensation image $I'(x, y)$ in this paper. Meanwhile, most nighttime images are the irradiation results of point light sources. To estimate the nighttime atmospheric light distribution more accurately, the atmospheric light estimation figure $A(x, y)$ of the degraded image, which computes by Equation (5), replaces the atmospheric light value A to estimate the transmittance. The final improved formula for calculating transmittance is shown in Equation (6) :

$$A(x,y) = \text{Gaussianfilter}\left(\max_{c\in\{r,g,b\}}\left(I^c(x,y)\right)\right) \quad (5)$$

$$t'(x,y) = \frac{1-\omega\cdot\left[I'(x,y)/A(x,y)\right]}{1-\omega\cdot\left[J'(x,y)/A(x,y)\right]} \quad (6)$$

where $t'(x, y)$ in Equation (6) denotes coarse transmittance. The atmospheric light estimation map $A(x, y)$ is usually taken as the average pixel value of the first 0.1 percent of the pixel values in descending order in the compensation image $I'(x, y)$.

However, the edge loss of the coarse transmittance $t'(x, y)$ is still serious, which is not suitable for subsequent dehazing. Joint bilateral filter can refine image texture and reduce image noise[7, 8], so it is used to further refine the coarse transmittance $t'(x, y)$. The processing flow is shown in Figure 3.

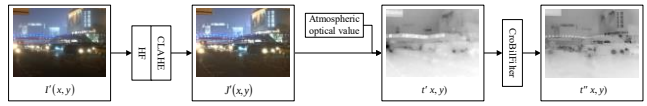

Figure 3. Flow chart of transmittance acquisition

2.3 Image Restoration

In the field of image dehazing, He et al analyzed outdoor images and the atmospheric scattering model proposed by Narasimhan et al[9]. They proposed the dark channel prior theory[5] that in the dark channel of outdoor images, more than 86 percent of pixel values are concentrated in the grayscale between 0 and 16. Assuming that atmospheric light value A is known, Zhang[10] deduces the mathematical expression of transmittance $t(x, y)$ in combination with dark channel prior theory, as shown in Equation (7) :

$$t(x,y) = 1-\omega[I^{dark}(x,y)/A] \quad (7)$$

where the parameter ω is the weight coefficient to maintain the realistic visual depth, which is taken as 0.95 in this paper. As transmittance $t(x, y)$ and atmospheric light value A are known, the reduction formula of the degraded image is deduced as shown in Equation (8).

$$J(x,y) = \frac{I(x,y)-A}{t(x,y)} + A \quad (8)$$

This algorithm has good dehazing performance in the experiments for daytime dehazing. But it has some limitations when applying on nighttime images dehazing. Therefore, a novel formula is proposed based on the refined transmittance $t''(x, y)$ and atmospheric light estimation figure $A(x, y)$ obtained in Section 2.2, which is shown in Equation (9).

$$J(x,y) = \frac{I'(x,y)-A(x,y)}{\max\left[t''(x,y),t_0\right]} + A(x,y) \quad (9)$$

where t_0 is the transmittance limit parameter added to prevent the mathematical calculation error when $t''(x, y)$ is 0. It is taken as 0.1 in this paper.

3. Experiments

3.1 Experiment Settings

All of the experiments in this paper are run on a computer with AMD Ryzen 7 4800H @ 2.90 GHz CPU, 16GB RAM, NVIDIA RTX 2060 Graphics card. The code is developed by C++ and the development platform is Visual Studio 2019. The size of the filter is set as 7 × 7. The dataset used in the experiments is from the night scene of Changan Avenue in Beijing under a heavy haze.

In order to verify the dehazing effect and runtime of the proposed algorithm in this paper, the experimental comparison is conducted from visual effect evaluation and objective evaluation. All the comparison algorithms used in this paper are classical dehazing algorithms.

3.2 Visual Effect Evaluation

In this section, two nighttime degraded images are randomly selected from the dataset and processed by different dehazing algorithms. The experimental results are shown in Figure 4. Figure 4(a) shows the original image and Figure 4(b) shows the result of Gao's[11] dehazing algorithm improved by dark channel prior theory, which demonstrates that it is not satisfactory to use daytime dehazing algorithm directly for nighttime dehazing while the

restored image scene is dark and the image information is seriously lost. Figure 4(c) shows the result of Xu's[4] algorithm based on the full-scale Retinex theory. It can be observed that the algorithm is effective. However, the brightness of the restored scene is low and there is still some loss of image texture. Figure 4(d) shows the result of Li's[12] algorithm for nighttime degraded images with multiple light sources. This algorithm is also effective for nighttime dehazing, yet the original color of the dehazing image tends to be cool and dark, which is not suitable for observation. Figure 4(e) presents the result of the proposed algorithm in this paper. Compared with other algorithms, the proposed algorithm in this paper has a good performance on the degraded area covered by the light source while effectively reduces the color distortion and texture loss.

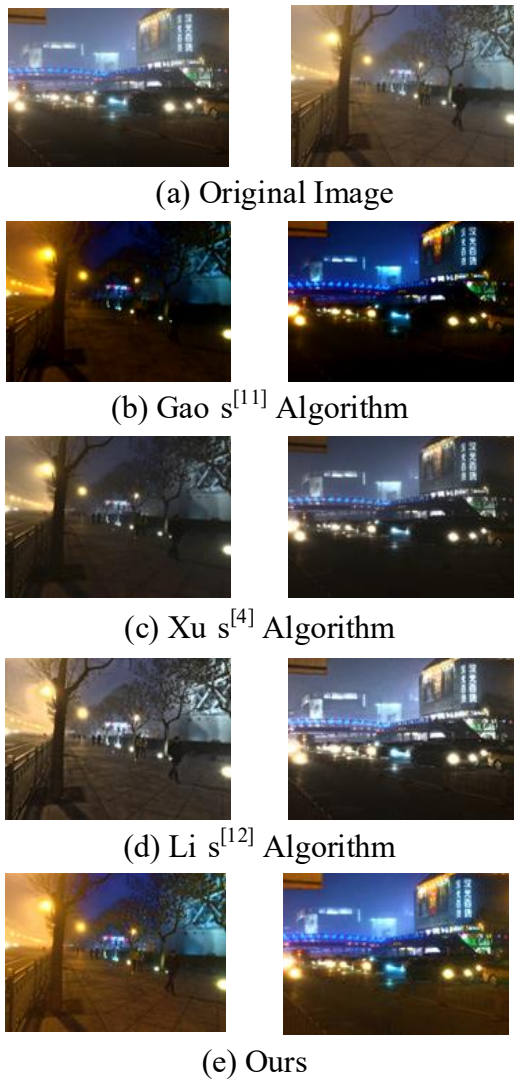

Figure 4. Visual effect comparison

3.3 Objective Evaluation

In order to further verify the effectiveness of the proposed dehazing algorithm in this paper, the objective evaluation criteria without reference are used to evaluate the performance of this algorithm. The objective evaluation criteria without reference include image contrast(c), mean gradients(g), information entropy(e), and algorithm running time(t). Contrast reflects the degree of gray contrast in an image[3]. The larger the value, the better the dehazing effect; The average gradient represents the tiny texture details in the image. The larger the value, the clearer the dehazing image is; Information entropy describes the amount of target feature information of the image. The larger the value, the more target information of the dehazing image.

The objective comparison of the experimental results in Figure 4 is shown in Table 1. Comparing the proposed algorithm in this paper with Gao's[11], it can be seen that it is not ideal for nighttime dehazing to apply the daytime dehazing algorithm directly. And the objective indexes of the dehazing image by using Gao's algorithm are lower than those of the original image, which is consistent with the conclusion drawn from the subjective evaluation in the visual effect. Compared with Xu's[3] and Li's[12] algorithms, the contrast, average gradient and information entropy of the proposed algorithm are increased by 28.74%, 27.15% and 5.61% respectively. Meanwhile, the runtime of the proposed algorithm is also shorter than others.

Table 1. Comparison of several objective evaluation results

Image	Method	*c*	*g*	*e*	*t*/s
Pic. 1 (600×431)	Original	54.76	4.42	7.35	-
	Gao's[11]	50.54	2.69	5.99	2.37
	Xu's[4]	49.19	3.57	6.77	4.01
	Li's [12]	83.15	5.36	7.24	12.88
	Ours	**95.35**	**6.78**	**7.43**	**1.54**
Pic. 2 (600×431)	Original	88.22	4.44	7.62	-
	Gao's[11]	86.25	4.61	5.78	2.66
	Xu's[4]	76.13	4.71	6.98	3.73
	Li's[12]	121.71	5.77	7.23	12.91
	Ours	**137.34**	**6.43**	**7.57**	**1.44**

4. Conclusion

In this paper, a nighttime dehazing algorithm based on adaptive global brightness compensation is proposed. Aiming at over-saturation of brightness compensation in existing algorithms, an adaptive global brightness compensation is proposed. The transmittance is obtained by homomorphic filtering, limited contrast adaptive histogram equalization algorithm and joint bilateral filtering, which effectively improves the accuracy of transmittance. Visual effect evaluation and objective evaluation both suggest that the performance of the algorithm is better than several similar dehazing algorithms. The proposed algorithm also effectively reduces the color distortion and texture loss of the restored image. It can be applied to the haze removal of nighttime degraded images of urban streets in the future.

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